# Hasan Mohd Hussain

*Machine Learning Engineer | Applied ML | Generative AI | Deep Learning & MLOps*
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## PROFESSIONAL SUMMARY

**Machine Learning Engineer** with an **MS in Computer Engineering**, building **Generative AI**, **RAG** and **agentic AI** systems. Cut QA time **95%** with **LLM evaluation** and automation; skilled in **Python**, **PyTorch**, **LangGraph**, **SQL**, **FastAPI** and **Docker**.

## EDUCATION

**University of Texas at Dallas** — **Richardson, Texas, USA**
Master of Science in Computer Engineering — Aug 2024 - Aug 2026
**GPA: 3.74/4.0** | *Relevant Coursework: NLP, Machine Learning, Computer Architecture, Embedded Systems*

**University of Mumbai** — **Mumbai, Maharashtra, India**
Bachelor of Engineering in Computer Engineering — Aug 2017 - Jul 2021
**GPA: 7.88/10.0** | *Relevant Coursework: Data Structures, Database Management Systems*

## SKILLS

**Machine Learning:** Supervised & unsupervised learning, deep learning, NLP, transformers, CNNs, computer vision, model evaluation, feature engineering, XGBoost, ensemble methods, time-series forecasting, statistical modeling, responsible AI, distributed systems
**Generative AI:** LLMs, RAG, embeddings, semantic search, prompt engineering, LLM-as-a-judge, tool calling, LoRA/PEFT fine-tuning
**Agentic AI:** LangGraph, LangChain, ReAct, multi-agent orchestration, Critic-Revise loops, human-in-the-loop, streaming, MCP
**Frameworks & APIs:** PyTorch, TensorFlow, scikit-learn, Keras, Hugging Face Transformers, Pandas, NumPy, OpenCV, spaCy, OpenAI API, Azure OpenAI, Gemini API, Ollama, AWS Bedrock
**MLOps & Deployment:** FastAPI, Flask, Docker, Kubernetes, AWS (SageMaker, S3, EC2), MLflow, CI/CD, model serving, monitoring, data pipelines
**Data & Vector DBs:** ChromaDB, FAISS, Pinecone, PostgreSQL, MongoDB, SQLite, SQL, Apache Spark
**Languages & Quality:** Python, SQL, Java, C++, Git, Linux, PyTest, Playwright, validation-based hyperparameter tuning, A/B testing, statistics

## WORK EXPERIENCE

**Mphasis** — **New York, United States**
Machine Learning Engineer — Jun 2026 - Aug 2026
- Engineered an **autonomous LLM-based QA agent** in **Python, Flask and Playwright** to evaluate a **production mortgage dashboard**.
- Designed **LLM-as-a-judge model evaluation** across **Claude, GPT-4o, Gemini and self-hosted Qwen**, cutting test time by **95%**.
- Created a **synthetic data generation pipeline** with answer keys for **5 financial document types**, yielding **200 labeled eval cases/run**.

**Mphasis** — **New York, United States**
Machine Learning Engineer — May 2025 - Aug 2025
- Built a **RAG pipeline** with **Azure OpenAI embeddings**, **ChromaDB** and **Gemini 2.5 Flash**, improving retrieval accuracy by **30%**.
- Implemented **semantic search** over loan PDFs with **Hugging Face (MiniLM) embeddings**, overlapping chunks and **page-level metadata**.
- Deployed the pipeline as **FastAPI REST endpoints** and optimized **Python backend logic**, improving **API response time 20%**.

**Dot9 Games** — **Mumbai, India**
Game Developer Intern — Apr 2024 - Aug 2024
- Built **Unity/C#** gameplay systems from design docs, raising **user engagement 25%** at a stable **60 FPS**, using **Git**.

**Shree Infotech** — **Mumbai, India**
Software Engineer — Apr 2023 - Mar 2024
- Developed **Python/Flask backend services** and **REST APIs** on **Docker**, automating workflows and saving **10+ hours/week**.
- Designed **MongoDB** schemas and indexes for **fast query retrieval**, and wrote **unit tests** (PyTest) to keep core services reliable.

**Nilee Games** — **Mumbai, India**
Game Developer Intern — Nov 2022 - Mar 2023
- Shipped a **2D physics-based driving game** in **Unity/C#** in **two weeks**: **500K downloads** in month one, **30%** higher engagement.

## PROJECTS

**FraudSight MLOps** | *Python, XGBoost, scikit-learn, SHAP, MLflow, FastAPI, Evidently*
- Architected **fraud detection** for **284,807** ULB transactions (**492** fraud): **XGBoost** vs. logistic baseline, **time-based splits**, **PR-AUC**.
- Specified an **MLflow** staging/production promotion gate, **FastAPI** scoring with **SHAP** explanations and **Evidently** drift retraining.

**NutriCoach.AI** | *Python, LangGraph, LangChain, ReAct, OpenAI, Gemini, Ollama, Flask SSE*
- Orchestrated a **7-agent LangGraph** pipeline with **6 ReAct tool-calling agents**, **11 tools** and **2 Critic-Revise feedback loops**.
- Streamed per-agent **Think-Act-Observe traces** via **Flask SSE**, generating **7-day meal plans** on **5 swappable LLM providers**.

**RAG Document QA System** | *Python, LangChain, ChromaDB, Hugging Face, Gemini*
- Developed a **5-stage RAG pipeline**: PDF ingestion, **500/100-char overlapping chunks**, **MiniLM embeddings** and **ChromaDB** storage.
- Integrated **Gemini** for **context-grounded QA** over **top-3 semantic retrieval**, tagging passages with source and chunk metadata.

## CERTIFICATIONS

**Anthropic:** Claude Code in Action, Claude Code 101, Claude 101 | **Udemy:** The Complete SQL Bootcamp
**IBM:** Machine Learning with Python, Building AI Agents and Agentic Workflows | **DeepLearning.AI:** Supervised ML, AI For Everyone